Vector Calculus Practice Questions (Thomas' Calculus Topics)

Green's Theorem

1. Let C be the positively oriented boundary of the region R enclosed by $y = x^2$ and $y = 4$. Evaluate $\oint_C (y^2)dx + (2xy)dy$ using Green's Theorem.

2. Verify Green's Theorem for $F(x, y) = (2x - y^2)i + (x^2 + 2y)j$ over the region bounded by $x^2 + y^2 = 1$.

3. Use Green's Theorem to compute the area of the region enclosed by $x^{2/3} + y^{2/3} = 1$.

Parametrization of Surfaces

4. Give a parametrization for the following surfaces:
   a. The part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the plane $z = 0$.
   b. The lateral surface of the cylinder $x^2 + y^2 = 9$ between $z = 0$ and $z = 4$.
   c. The portion of the cone $z = \sqrt{x^2 + y^2}$ between $z = 0$ and $z = 2$.

5. Find a parametrization for the sphere $x^2 + y^2 + z^2 = a^2$ and compute the corresponding tangent vectors r_θ and r_φ.

Surface Area

6. Find the surface area of the part of the plane $3x + 2y + z = 6$ that lies in the first octant.

7. Find the area of the part of the paraboloid $z = 1 - x^2 - y^2$ that lies above the xy-plane.

8. Compute the surface area of the part of the sphere $x^2 + y^2 + z^2 = 9$ that lies above the plane $z = 2$.

Scalar Surface Integral

9. Evaluate $\iint_S x \, dS$ where S is the part of the plane $x + y + z = 1$ in the first octant.

10. Compute $\iint_S z^2 \, dS$ where S is the portion of the cylinder $x^2 + y^2 = 4$ between $z = 0$ and $z = 3$.

Divergence Theorem

11. Verify the Divergence Theorem for $F(x, y, z) = (x^2, y^2, z^2)$ over the region bounded by the cube $0 \leq x, y, z \leq 1$.

12. Use the Divergence Theorem to compute the outward flux of $F(x, y, z) = (yz, zx, xy)$ across the surface of the sphere $x^2 + y^2 + z^2 = a^2$.

13. Let $F = (x + y, y + z, z + x)$. Use the Divergence Theorem to find $\iint_S F \cdot n \, dS$ where S is the boundary of the solid bounded by $z = 0$ and $z = 4 - x^2 - y^2$.

Stokes' Theorem

14. Let $F(x, y, z) = (y, -x, z)$ and S be the upper hemisphere $x^2 + y^2 + z^2 = 1$, $z \geq 0$. Use Stokes' Theorem to compute $\iint_S (\nabla \times F) \cdot dS$.

15. Verify Stokes' Theorem for $F(x, y, z) = (z, y, x)$ and S is the portion of the plane $x + y + z = 1$ above the triangle bounded by the coordinate planes.

16. Compute the circulation of $F = (y^2, z^2, x^2)$ around the triangle with vertices (1,0,0), (0,1,0), (0,0,1) using Stokes' Theorem.